

ESE 499 Undergraduate Research –Summer 2026

Proposal

**Comparative Analysis of Environment-Aware Resilient Wireless
Communication Links in Disaster Scenarios Using SDR**

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1. Introduction & Motivation

In extreme disaster scenarios, conventional communication infrastructure is often severely compromised or destroyed, creating critical bottlenecks for search, rescue and recovery operations. To overcome this, highly resilient wireless communication systems, such as ad-hoc networks or UAV-assisted relays, are essential. However, the propagation behavior of RF signals differs drastically depending on the operational environment—whether in a highly dense urban area like Manhattan or a suburban environment like Stony Brook. This research aims to mathematically model the signal behavior in these contrasting environments and validate the theoretical findings using Software-Defined Radio (SDR) hardware, ultimately proposing environment-aware communication recovery strategies.

2. Proposed Methodology

The research will be conducted in two main phases: theoretical simulation and empirical hardware validation

- Phase 1: Channel Modeling and Simulation (MATLAB & Python)

- **Dense Urban Environment** : To account for obstructed Line of Sight and severe multipath reflections, Rayleigh Fading will be applied alongside Log-normal Shadowing with a high variance.
- **Suburban Environment** : Given the probability of a partial LOS, Rician Fading will be modeled.
- A generalized path loss model will be utilized to analyze the signal attenuation over distance d and evaluate the Signal-to-Noise Ratio (SNR) and Bit Error Rate (BER) across both environments:

$$PL(d) = PL(d_0) + 10n\log_{10}\left(\frac{d}{d_0}\right) + X_\sigma$$

- Phase 2: Hardware-in-the-Loop SDR Verification

- PlutoSDR and USRP B210 hardware will be utilized to empirically validate the simulation baseline in real-world settings.
- Custom transceiver chains will be designed using GNU Radio. Leveraging my previous experience in UAV communication modeling using Python and C++, I will analyze practical RF parameters, including signal attenuation, distance-based degradation, and overall system reliability under varying constraints.

-Phase 3: Data-Driven Environment-Aware Algorithm Design & Synthesis

- **Empirical Data Analysis & Channel Profiling:** Conduct a rigorous statistical analysis of the RSSI, SNR, and BER datasets collected during the summer session (Phases 1 and 2) across urban (Manhattan) and suburban (Stony Brook) environments. This involves establishing exact communication degradation thresholds and calibrating the theoretical path loss and fading models based on real-world empirical data.
- **Adaptive Wireless Communication Algorithm Derivation:** Develop a mathematical algorithm designed to maximize link survivability under infrastructure-compromised disaster scenarios. Leveraging the analyzed channel metrics, an Adaptive Modulation and Coding (AMC) framework or an environment-aware dynamic frequency selection mechanism will be



designed, enabling the system to autonomously optimize transmission parameters based on real-time Channel State Information (CSI).

- **Manuscript Compilation in IEEE Paper Format:** Evaluate the performance of the proposed adaptive strategy against conventional static baselines in terms of bit error rates and link reliability. Synthesize the theoretical derivations, empirical curves, and algorithmic flowcharts into a formal, publication-ready final report structured strictly according to the IEEE Conference Paper Template.

3. Expected Outcomes & Timeline

To efficiently manage time and physical resources, this research is structured across the summer session and the early fall semester:

-Dedicated Summer Lab Hours:

I will dedicate **Mondays from 2:00 PM to 4:30 PM** and **Wednesdays from 12:00 PM to 2:30 PM** strictly to working in the lab, focusing entirely on this research project.

-Summer Session (June 1 – Aug 15): Simulation & Data Collection

Focus on the time-intensive tasks: completing Phase 1 (simulation modeling) and Phase 2 (SDR testbed setup and field data collection) during the dedicated lab hours to secure all necessary empirical data before the fall semester.

-Early Fall Semester (Aug 24 – Sep 30): Algorithm Design & Paper Drafting

Utilize the collected data to finalize Phase 3 (designing the environment-aware communication strategy).

Synthesize the research findings and develop a comprehensive final report **written strictly in the IEEE Paper Format** to finalize the research as a formal, publication-ready manuscript by the end of September.

